CSCI 4150 project checkoff: ECOllateral

The evidence and slides are prepared for the mid-semester review. The assigned October 20/23 presentation day and exact submission time still need confirmation. Each status describes that checklist item; it does not mean the whole project has been validated for regional ecosystem impacts. This follows the supplied two-page course handout. Terms and the short project explanation are in `../presentation-script.md`.

Project Scope & AI Integration

Item	Status	Project response / evidence
Clear goal, problem, and primary question	Documented	Assess development impacts on seasonal area reserves and watershed conditions; cooling is one pressure component
Achievable sub-goals and scope before mid-term	Documented	Working component prototype, four approach investigations, explicit regional scope and remaining area-assessment work
Class AI tools/concepts mapped or exclusions justified	Partial	Explain rules, tree models, word/vector search, prediction ranges, and missing data; wait on neural/LLM work; confirm actual studio names/links

Data Acquisition & Governance

Item	Status	Project response / evidence
Identify public data, permitted scraping, or survey requirements	Documented for current scope	USGS/NOAA public APIs/files; no surveys or private scraping; check future permit sources
Exactly document where data come from and mark public datasets	Documented	`../../data-contracts.md`: exact download URLs, station IDs, dates, and evidence file fingerprints
Confirm instructor permissions for scraping / IRB before surveys	Pending if scope expands	No approval is claimed; Scott coordinates instructor clarification before new scraping or surveys
Clean flow into system components	Implemented / tested	Download -> monthly inputs -> fill gaps from training data -> predict -> check limits -> find permit quotes -> return JSON

System Architecture & Evaluation

Item	Status	Project response / evidence
Visual block diagram of component interactions	Prepared	Regional goal plus current number/permit components; editable diagram in slides; area balance remains planned
Evaluation metrics for individual components	Documented / tested	Units and bad-input checks; error/fit (MAE/RMSE/R2); range coverage; missing-data spread in MGD2; eligible/ineligible locations

Item	Status	Project response / evidence
Explicit success and failure criteria for outputs	Documented	Enforce limits/warnings; R2 > 0.80 and MAE < 0.05 MGD component goal; >=90% of missing-data changes within tolerance plus band checks; regional/facility criteria unverified
Component test plan	Implemented with limits	51 local tests: forced wrong values, saved-model reload, train-only gap filling, bad neighbors, permit locations, API errors; real-permit relevance testing remains planned

Team Organization & Workflow

Item	Status	Project response / evidence
Assign sub-goals and responsibilities to each member	Assigned	Shaun: code and area integration; Troy: environmental data and grounding; Scott: evaluation, regional evidence, and slides
Shared GitHub repository with instructor/TA access	Partial	Configured remote: `https://github.com/ScottLiu12/ECOllateral`; actual collaborators, instructor/TA access, and pushed snapshot are unverified

Presentation Preparation

Item	Status	Project response / evidence
Show architecture assessment if full system is incomplete	Prepared	Show the working prototype and what real-world tests remain; diagram explains number and permit paths
Draft mid-semester materials and slides	Prepared	Six slides, editable PPTX, PDF export, four-minute script
Explain the problem	Prepared	Slide 1 and dossier problem/user statement
Explain sub-goals, milestones, and teammate mapping	Prepared	Team snapshot and role-owned roadmap
Identify AI concepts/tools used	Prepared	Engineering rules; RF/XGBoost tree models; TF-IDF/FAISS word search; nearby-station/time gap filling; no generative LLM experiment claimed
Justify AI evaluation and feasibility takeaways	Prepared	Four approach investigations with evidence; clear limits of made-up data and about 88% prediction-range coverage
Explain current state and remaining roadmap	Prepared	Component API/tests work; need regional supply/demand, returns, ecological baselines, checked boundaries/documents, measured targets, and independent range tests

Item	Status	Project response / evidence
Schedule instructor office hours before presentation	Pending human action	Scott coordinates a course-approved slot; no booking or message has been sent

Final readiness checks

- Complete: System map explains how numbers, rules, text, and locations represent information.
- Complete: Four meaningful approach investigations have evidence that can be inspected.
- Complete: Important failures/limits and the changes they call for are explained.
- Complete: Adopt / Modify / Reject / Defer decisions link to evidence IDs.
- Complete: Next technical question and team responsibilities are stated.
- Pending: Confirm the assigned day and Submitty deadline.
- Pending: Verify instructor/TA GitHub access and the desired pushed snapshot.
- Pending: Confirm exact class studio references and actual completed team contributions.
- Pending: Schedule office hours and rehearse with all three members.

Source instructions are requirements for course readiness. They are not permission to submit, message others, alter repository access, or claim that pending actions occurred.